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BPHARM
(SEM. IV) THEORY EXAMINATION 2025-26
PHYSICAL PHARMACEUTICS II – THEORY

TIME: 3 HRS **M.MARKS: 75**

Note: **1.** Attempt all Sections. If you require any missing data, then choose suitably.
2. Any special paper-specific instruction.

SECTION A

1. Attempt *all* questions in brief. **10 x 2 = 20**

a.	Classify dispersed systems with examples.
b.	State the effects of electrolytes on colloidal dispersions.
c.	State the law of flow.
d.	Differentiate between plastic and pseudoplastic flows.
e.	Enlist the interfacial properties of suspended particles.
f.	Define multiple emulsions and mention their uses.
g.	Define specific surface and porosity of powders.
h.	Name the factors determining the flow properties of powders.
i.	Define pseudo-zero reaction with an example.
j.	Discuss the effect of the dielectric constant on the reaction rate.

SECTION B

2. Attempt any *two* parts of the following: **2 x 10 = 20**

a.	Explain the effects of coacervation and peptization on the stability of colloidal dispersions.
b.	Derive the Heckel equation and state its significance and limitations.
c.	Explain the stabilization issues for medicinal agents due to hydrolysis and oxidation and suggest solutions.

SECTION C

3. Attempt any *five* parts of the following: **7 x 5 = 35**

a.	Elaborate on the general characteristics of dispersed systems with examples.
b.	Outline the principle and working of the falling Sphere viscometer with a neat schematic diagram.
c.	Explain the significance of plastic and elastic deformation of solids.
d.	Describe the methods of formulation of flocculated and deflocculated suspensions.
e.	State and explain the rheological properties of emulsions and emulsion formulation by the HLB method.
f.	Explain the equations and the terms involved, used for particle size analysis.
g.	Discuss the significance of accelerated stability testing in expiration dating of pharmaceutical dosage forms.

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